MediFlow AI — User Stories

How to read this

Every feature in MediFlow is written here as user stories, in the order a patient moves through the clinic: consultation, guidelines, care plan, follow-up, messages, claim. Each story has acceptance criteria, which are the checks a coding agent must pass before the story counts as done.

Format: *As a* [role], *I want* [action], *so that* [benefit].

Roles: Doctor, Nurse, Front desk, Billing officer, Patient, Admin (the builder running demos and evaluations).

Priority: Must = needed for the demo to work end to end. **Should** = makes the demo stronger; build if time allows.

Epic 1: Clinical note (Scribe)

ID	Story	Acceptance criteria	Priority
US-1.1	As a doctor, I want to see today's consultations in one list, so that I know which notes still need writing.	List shows all transcripts with patient name, date and note status (none, draft, needs attention, approved).	Must
US-1.2	As a doctor, I want to generate a draft note from a transcript in one click, so that I don't write it from scratch.	Draft appears in under 30 seconds with all 13 fields; LLM-filled fields are marked "AI draft".	Must
US-1.3	As a doctor, I want the transcript next to the draft, so that I can check every field against what was said.	Side-by-side layout; transcript scrolls independently.	Must
US-1.4	As a doctor, I want doubtful fields highlighted with a reason, so that I know where to look first.	Fields failing validation show in amber with the reason (e.g. "code format invalid").	Must
US-1.5	As a doctor, I want to edit any field and approve the note, so that the final record is mine.	Approve is disabled while a required field is empty; on approve, status becomes approved and the edited fields are recorded.	Must

ID	Story	Acceptance criteria	Priority
US-1.6	As a doctor, I want to regenerate a poor draft, so that I don't have to fix a bad one by hand.	Regenerate re-runs the pipeline and replaces the unapproved draft.	Should
US-1.7	As a patient, I want my identity hidden from the AI service, so that my details don't leave the clinic.	Text sent to the LLM contains no real name, phone or NHIS number (verified in logs); real names reappear on the review screen.	Must
US-1.8	As a doctor, I want the note to open even when the AI fails, so that I can still work.	On AI failure, the form opens empty with "AI unavailable, fill manually".	Must

Epic 2: Guidelines (Protocol assistant)

ID	Story	Acceptance criteria	Priority
US-2.1	As a doctor, I want to check my prescription against the clinic guidelines before I approve, so that I catch mistakes early.	Check guideline marks each drug as matches, dose differs, not first-line or not in guideline, with a citation; the 2 planted mismatches are flagged.	Must
US-2.2	As any staff member, I want to ask the guidelines a question in plain words, so that I don't have to search the PDFs.	Answer in under 5 seconds, every statement cited by document, section and page.	Must
US-2.3	As any staff member, I want to open the source behind a citation, so that I can trust the answer.	Clicking a citation shows the original chunk text.	Must
US-2.4	As any staff member, I want a clear "not covered" answer when the guidelines don't say, so that I'm never given a made-up answer.	All 5 out-of-scope test questions return the refusal sentence; the LLM is not called on weak matches.	Must
US-2.5	As an admin, I want to re-load the guideline documents with one click, so that answers stay current.	Re-ingest replaces chunks and shows the chunk count and time.	Should

Epic 3: Care plan (Care coordinator)

ID	Story	Acceptance criteria	Priority
US-3.1	As a nurse, I want a care plan created automatically when a doctor approves a note, so that I don't plan each step by hand.	Approving a note starts a run; the plan reaches "awaiting approval" in under 30 seconds.	Must
US-3.2	As a nurse, I want every proposed action to show its reason and source, so that I know why it was suggested.	Each action shows a reason and a note field or guideline citation.	Must
US-3.3	As a nurse, I want to approve, edit or reject each action, or approve all, so that nothing happens without me.	Actions only execute after approval; rejected actions are kept with a reason.	Must
US-3.4	As a nurse, I want disagreements between the note and the guideline flagged, so that the agent never picks a side for the doctor.	The conflict test case shows a flag at the top and no auto-chosen date.	Must
US-3.5	As a nurse, I want to see the agent's step-by-step trace, so that I can check its reasoning.	Trace lists every step: reasoning line, tool, input, result, time.	Must
US-3.6	As a doctor, I want a referral letter drafted only when the guideline calls for it, so that patients aren't referred needlessly.	Letter drafted for the referral test case and for no other case.	Should
US-3.7	As an admin, I want to switch the coordinator off, so that the clinic still works if the agent fails.	With the coordinator off, scoring, messages and claims run on the direct triggers.	Should

Epic 4: Appointments and no-show risk

ID	Story	Acceptance criteria	Priority
US-4.1	As front desk, I want every new follow-up to get a no-show risk score automatically, so that I know who may not come.	Approved follow-ups are scored on creation with band High, Medium or Low.	Must
US-4.2	As front desk, I want a one-sentence reason for each risk, so that I understand and can act on it.	Each score shows a plain sentence naming only its top 3 factors; template sentence used if the AI fails.	Must
US-4.3	As front desk, I want upcoming appointments sorted by risk, so that I focus on the riskiest first.	Schedule sorts by score, highest first, with reminder and confirmation status.	Must
US-4.4	As front desk, I want to mark each appointment attended or missed, so that we measure results and improve the model.	Attended or Missed saves to the appointment and appears in results.	Must
US-4.5	As an admin, I want the model's quality shown against a simple baseline, so that I can prove it adds value.	Model card shows AUC, precision and recall for both models; the chosen model beats the baseline.	Must
US-4.6	As an admin, I want to check that the model doesn't rely on sex, so that it's fair.	Model card shows sex near the bottom of SHAP importance.	Should

Epic 5: Patient messaging

ID	Story	Acceptance criteria	Priority
US-5.1	As a patient at high risk of missing a visit, I want reminders before my appointment, so that I don't forget.	High risk: 2 reminders (3 days and 1 day before); Medium: 1; Low: none. Reminders never mention the diagnosis.	Must
US-5.2	As a patient, I want to confirm or change my appointment by replying, so that I don't have to call.	"1" confirms; "2" offers 3 free slots; choosing one rebooks and frees the old slot.	Must

ID	Story	Acceptance criteria	Priority
US-5.3	As a patient, I want plain-English instructions after my visit, so that I know how to take my medicines and when to come back.	Message covers what was found, medicines, danger signs and next visit; readability score ⩾ 60.	Must
US-5.4	As a nurse, I want to approve after-visit instructions before they're sent, so that patients only get correct advice.	Messages wait for approval; any medicine name, dose or duration that doesn't match the note blocks sending.	Must
US-5.5	As a patient, I want simple questions (hours, directions, what to bring) answered quickly, so that I get help without waiting.	Logistics questions are answered from the guideline documents; nothing is invented.	Should
US-5.6	As a nurse, I want every medical question sent to me rather than answered by the AI, so that patients never get unsafe advice.	All clinical test messages reach the nurse inbox; the patient gets the hand-off message.	Must
US-5.7	As a patient in an emergency, I want an immediate, clear instruction, so that I get help fast.	Emergency words trigger the emergency reply and a red staff alert without calling the AI; 6 of 6 test emergencies caught.	Must
US-5.8	As a patient, I want to stop messages by replying STOP, so that I control what I receive.	STOP opts the patient out; no further reminders; START turns them back on.	Must
US-5.9	As a patient, I want no messages at night and not too many in a day, so that I'm not disturbed.	No sends from 20:00 to 07:00 (except emergency replies); maximum 2 automated messages a day.	Should

Epic 6: NHIS claims

ID	Story	Acceptance criteria	Priority
US-6.1	As a billing officer, I want each claim built automatically from the approved note, so that I don't re-type diagnoses and medicines.	Claim lines come from the note; quantities and prices come only from the tariff table; totals add up.	Must

ID	Story	Acceptance criteria	Priority
US-6.2	As a billing officer, I want claims sorted red, amber and green, so that I fix the riskiest first.	Queue orders red (blocking errors), amber (warnings or AI queries), then green.	Must
US-6.3	As a billing officer, I want each problem explained with a suggested fix, so that I can correct it quickly.	Every flag shows the check, a reason and, where possible, a fix (e.g. correct code or price).	Must
US-6.4	As a billing officer, I want the AI to question drugs that don't fit the diagnosis or notes that don't support it, so that I catch errors rules can't see.	LLM queries cite a guideline chunk; queries without a citation are dropped.	Must
US-6.5	As a billing officer, I want to override a warning with a written reason, so that I can submit valid exceptions.	Amber flags can be overridden only with a reason, saved with my name; red flags cannot be overridden.	Must
US-6.6	As a billing officer, I want to submit a clean claim and see the result, so that I know if it will be paid.	Submit returns paid or rejected with reasons; rejected claims return to the queue.	Must
US-6.7	As a clinic manager, I want to see the claim rejection rate and amount paid, so that I know the money impact.	Results show the rejection rate before vs after checks, error catch rate and GH₵ recovered (synthetic data).	Should

Epic 7: Admin, results and demo

ID	Story	Acceptance criteria	Priority
US-7.1	As an admin, I want to switch between roles from the top bar, so that I can demo every screen alone.	Role switcher changes the menu and permissions instantly.	Must
US-7.2	As an admin, I want to reset all demo data with one click, so that every demo starts clean.	Reset rebuilds patients, transcripts, appointments and claims from the seeded generators.	Must

ID	Story	Acceptance criteria	Priority
US-7.3	As an admin, I want to run the daily job for any date, so that I can show reminders without waiting days.	"Run daily job for date" sends the reminders due on that date.	Must
US-7.4	As an admin, I want one results page across all modules, so that a reviewer sees the evidence in one place.	Results page shows every KPI against its target, with model and prompt versions and run date.	Must
US-7.5	As a reviewer, I want to see that every AI action and approval is logged, so that I trust the system is auditable.	Audit log lists who or what acted, on what, with which model and prompt version.	Should
US-7.6	As any user, I want a clear notice that this is a prototype on synthetic data, so that nobody mistakes it for a clinical tool.	Banner shows on every screen.	Must

Coverage

48 stories across 7 epics: 39 Must and 9 Should. Ama's full journey (US-1.2 → US-1.5 → US-2.1 → US-3.1 → US-3.3 → US-4.1 → US-5.1 → US-5.3 → US-6.1 → US-6.6) uses only Must stories, so the end-to-end demo can be completed before any Should story is started.